

Germanium dioxide nanocubes with different size distributions and its Gibbs free energy theory

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Abstract: A simple synthesis method for single crystal GeO₂ nanocubes in a reverse micelle system was reported here. Hydrolysis of germanium tetrachloride (GeCl₄) in a micelle system produces GeO₂ nanocubes in the presence of oleylamine and cetyltrimethylammonium (CTAB). We found that we could obtain GeO₂ nanocubes with monodisperse size distribution, bimodal size distribution and polydisperse size distribution by varying the reaction parameters. We put forward a feasible explain for this phenomenon with using Gibbs free energy theory, and this theory can give some meaningful guidance for preparing monodisperse nanomaterials.

Introduction

Exact control of size, distribution and morphology of nanomaterials is an important task in the design of advanced materials due to the size dependent physical and chemical properties of nanoparticles which can be significantly different from the corresponding bulk materials. For example, nanometer-sized semiconductors have electronic and optical properties that depend on the particle size, making them potential candidates for applications where tunability of these properties is essential [1-3]. Size distribution is one of the major issue in current research activities in nanotechnology, for example, for silica materials, researchers reported that the occurrence of pore with a bimodal pore size distribution in mesoporous materials is important and useful for catalysis and for the engineering of pore system [4-5]. Many different processes, such as chemical vapor deposition, etching, milling, biomineralization, and so forth, are currently under investigation for the production of nanomaterials with different size distribution [6-8]. The microemulsion precipitation is known for its ability to synthesize narrowly distributed nanomaterials with a defined size. The main benefit of this technique results from the nanosized droplets (reverse micelles, disperse water droplets in the continuous oil phase), which limit the possible amount of reactants to be stored inside one droplet. Thus, always well-defined conditions for nucleation and growth are provided in the system. Additionally, adsorbed molecules from the surfactant shells of the droplets can act as growth directors to form complex particle morphologies when the particle dimension reaches the boundaries of the nanoscale cavity. Further, the surfactant shell also protects the nanoparticles against agglomeration to larger particles or breakage into smaller particles [4].

GeO₂ has many attractive properties: it is a dielectric oxide and a blue photoluminescence material with peak energies around 2.2 and 3.1 eV, and it has a refractive index and a linear coefficient of thermal expansion that are higher than for silicate (SiO₂). Moreover, it is a promising material for optical wave guides and nanoconnections in optoelectronic communication and vacuum technology [9]. Recently, much attention has been paid to the preparation of nanostructures of germanium dioxide (GeO₂), including nanowhiskers [10], nanowires [11], nanorods [12], nanosheets [13], nanofibers [14] and nanocubes [15]. Here we report the preparation of germanium dioxide nanocubes with different size distributions by reverse micelle system, we could get monodisperse, bimodal and polydisperse distributions by varying experiment parameters. We established a feasible theory to explain the phenomena based on the Gibbs free energy, which also can give us useful guidance for preparing high quality monodisperse nanomaterials.

Experimental details

To produce GeO_2 nanocubes, germanium tetrachloride (GeCl_4), heptane (or octane), and water were used as precursor, oil phase and water phase, respectively. Oleylamine and cetyltrimethylammonium (CTAB) were used as surfactants. Two types of micelle solutions were prepared: (1) a 3 ml aqueous solution with pH=1.0, 5 g CTAB and 9 ml pentanol were added to 60 ml heptane to form a transparent microemulsion; (2) 2 ml of a 0.5 M GeCl_4 tetrahydrofuran (THF) solution and 5 ml oleylamine were added to 20 ml heptane to form a transparent microemulsion. The GeCl_4 (THF)-oleylamine-oil phase solution was dropped into the water-CTAB-oil phase solution accompanied with strong magnetic stirring. The solution becomes turbid after 20 min, and the reaction is complete after 180 min. Subsequently, the particles were separated from the solution by centrifugation at 10^4 rpm for 20 min. The particles were then redispersed in ethanol in an ultrasonic bath. The centrifugation was repeated twice so as to remove all of the surfactant. In the final product, the precipitant containing nanocubes with white color was collected and redispersed in a small amount of xylene for further characterization. Changing the concentration of GeCl_4 from 0.5M to 1.2M in solution, we can get GeO_2 nanocubes samples with different size distributions. X-ray powder diffraction (XRD) was performed with a Rigaku D/MAX-2550PC powder X-ray diffractometer in Bragg-Brentano geometry, using Cu K_α radiation. The diffraction pattern was collected in the 2θ range 15-80 degrees in steps of 0.02 degree and counting time 2s per step. The crystallite size was calculated from the broadening of the diffraction peak at $2\theta = 26^\circ$ using the Scherrer equation. Nanocrystal size, shape and crystallography were characterized by transmission electron microscopy (TEM) using a JEOL TEM CX200 microscope operated at 160 kV. TEM samples were prepared by dropping the liquid containing GeO_2 nanoparticles on a carbon coated copper substrate.

Results and discussion

With varying the concentration of precursor (GeCl_4) in solution, we can obtain GeO_2 with different sizes distributions as shown in Fig.1. From it we can see that the size distribution changes from monodisperse to polydisperse with increasing the concentration of GeCl_4 from 0.5M to 1.2M. Figure 1b shows us the products with bimodal size distribution, the smaller size is $4 \pm 0.4 \text{ nm}$ and the bigger size is $9 \pm 0.6 \text{ nm}$, this kind of size distribution also have been observed by other researchers on other materials [17-18].

Fig.2 shows the XRD patterns of GeO_2 with different size distribution, the average size of the nanocubes, as estimated from the Scherrer equation, are 8, 9.5 and 13 nm, which are consistent with the TEM observations. The diffraction peaks are attributed to the (100), (101), (110), (102), (200), (112), et al. planes of GeO_2 , the smaller edge length of nanocubes, the broader of the peaks. It appears that the larger nanocubes have relatively higher crystallinity.

Dispersions such as emulsions and foams are sensitive to coarsening phenomena like coalescence and Ostwald ripening, since their thermodynamically most stable state is the completely demixed one. Coalescence is often considered as the most important destabilization mechanism leading to coarsening of dispersions. However, coalescence can often be prevented by a careful choice of stabilizers and is mainly of interest during processing. On the other hand, Ostwald ripening will continuously occur as soon as curved interfaces are present. The curvature of particles causes higher solubilities of the dispersed phase at the particle boundary compared to in the bulk or near to larger particles. The concentration gradient of the dispersed phase in the continuous phase causes large particles to grow at the expense of smaller particles. There are many reports about the size distribution of products, including using some chemical or physical methods to do size choosing process [16]. Such as in reference [17], the authors observed that CeO_2 nanocubes with bimodal size distribution can be obtained at the high concentration of Ce (III) in the water phase (50.0 mmol/L). Large-sized and small-sized NPs coexist in the whole reaction system in this situation, whereas the large-sized nanocubes could not be further smoothed at the cost of the small particles. This implies that the current system may be absent of Ostwald ripening because of the low

solubility of the OLA-capped ceria in the water phase. The larger size of CeO_2 nanocubes increased with increasing the concentration of Ce^{3+} in system. Furthermore, in reference [18], the authors also considered that Ostwald ripening can be excluded due to the coexistence of two particle populations with a different mean size in the final state of experiment, and thought that coagulation seemed to be the most probable mechanism for the formation of larger particles if the surfactant shells are destabilized, although no agglomerates were observed in the TEM pictures. Dissolution of the small particles which would be necessary for Ostwald ripening is not observed.

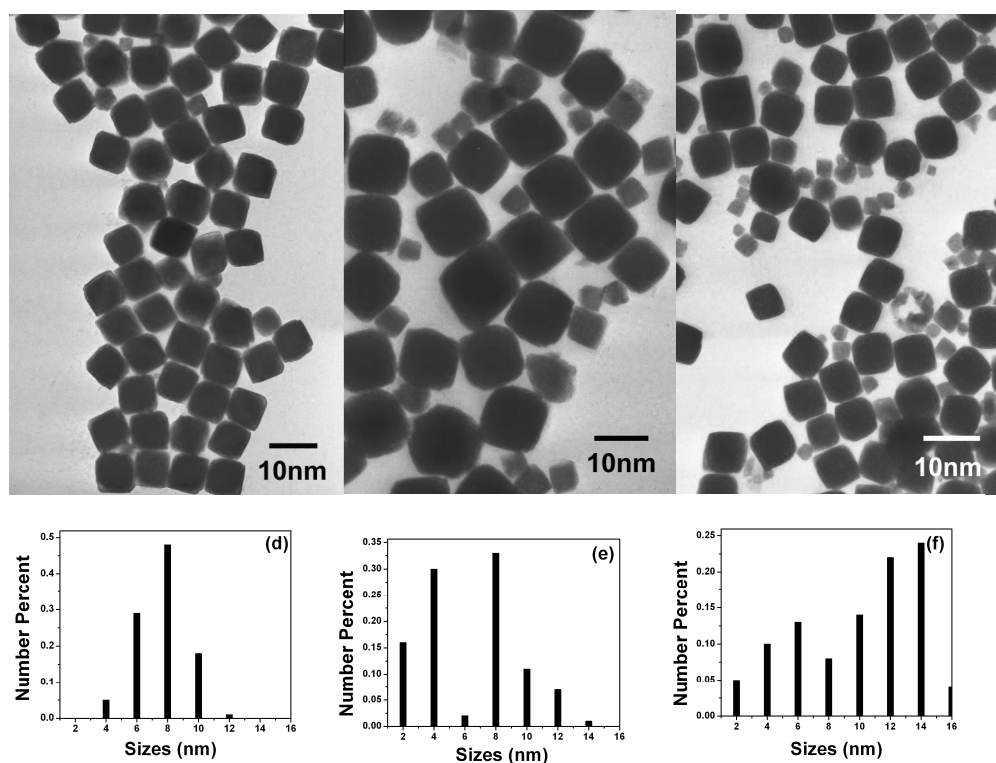


Figure 1 From (a) to (c), TEM images for GeO_2 nanocubes with monodisperse, bimodal size and polydisperse distributions, the precursor concentration for GeCl_4 is 0.5M, 0.8M and 1.2M, respectively. From (d) to (f), size distribution columns for samples from (a) to (c), respectively.

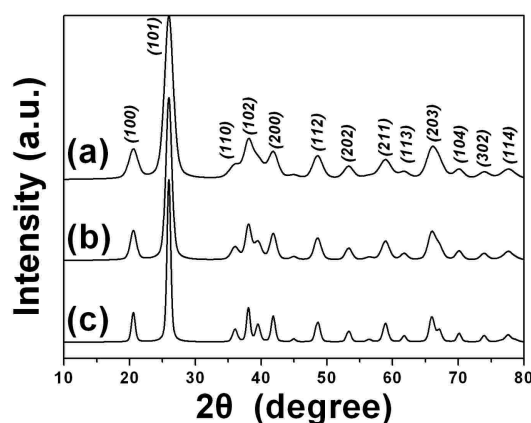


Figure 2 XRD patterns for GeO_2 nanocubes obtained with different GeCl_4 concentrations. From (a) to (c), the concentration for GeCl_4 is 0.5M, 0.8M and 1.2M, respectively.

In current station, we can get GeO_2 with different size distributions when we varying the precursor concentration in experimental. We think the coexist of bigger nanocubes and small nanocubes can be explained by the free energy of the system. The system (the nanocubes) can

present different size distribution when the free energy is different. Considering the depiction, we use bimodal size distribution as an example to explain the coexisting phenonema.

Before deducing, four simplified assumptions are put forward: a) the considered product is M mol and the volume is V_{mol} (m^3); b) if we obtained nanocubes with two kinds sizes, the lengths of side are d_1 (m) and d_2 (m), the amounts are n_1 and n_2 , respectively; (c) if we obtained nanocubes with homogeneous size, $d_1 = d_2 = d_0$ and the amount is n_0 .

The system changes from a bimodal size distribution to monodisperse distribution, the corresponding changes of entropy, enthalpy and Gibbs free energy are:

$$\Delta S_{mix} = k \left(n_1 \ln \frac{n_1 + n_2}{n_1} + n_2 \ln \frac{n_2 + n_2}{n_1} \right)$$

$$\Delta H = \gamma (n_1 d_1^2 + n_2 d_2^2 - n_0 d_0^2)$$

$$\Delta G = \Delta H - T \Delta S_{mix}$$

γ is proportional to the surface energy of GeO_2 and related with the morphology of the particles; k is Boltzmann constant;

$$n_1 d_1^3 + n_2 d_2^3 = V_{mol} = n_0 d_0^3 \quad (1)$$

The core is to obtain the minimum value of ΔG under satisfied with the equation (1). We have to lead-in Lagrange constant α and function

$$F(d_1, n_1, d_2, n_2, \alpha) = \Delta H - T \Delta S_{mix} + \alpha (n_0 d_0^3 - n_1 d_1^3 - n_2 d_2^3)$$

Then we discuss the effect of n_1 and n_2 on the F :

$$\frac{\partial F}{\partial n_i} = \gamma d_i^2 - kT \ln \left(\frac{n_1 + n_2}{n_i} \right) - \alpha d_i^3 \quad (2)$$

$$\frac{\partial^2 F}{\partial n_i^2} = kT \frac{n_1 + n_2 - n_i}{n_1 + n_2} > 0 \quad (3)$$

When $\frac{\partial F}{\partial n_i} = 0$ $i = 1, 2$, we can obtain the minimum value of F :

$$-\frac{\gamma}{kT} d_1^2 d_2^2 (d_2 - d_1) + d_2^3 \ln \left(1 + \frac{n_2}{n_1} \right) - d_1^3 \ln \left(1 + \frac{n_1}{n_2} \right) = 0 \quad (4)$$

Based on equation (4), we can get a series of relation between d_1 and d_2 with ΔG .

Figure 3 shows the relationship of Gibbs free energy of the system with changing of d_1 and d_2 . The temperature is 500K, and the surface energy

$$\gamma = 7.76 \times 10^{-4} J / m^2$$

$$V_{mol} = 10^{-2} cm^3, \quad k = 1.38 \times 10^{23}$$

Line 1 illustrates the value of d_2 when the system energy reaches to the maximum Gibbs energy under different d_1 , and line 2 illustrates the value of d_1 when the system energy reaches to the maximum Gibbs energy under different d_2 . From Fig. 3 we can see when d_1 is smaller than 6 nm,

the system Gibbs free energy is decreased with increasing d_2 , which can be used to explain the phenomenon that two sizes of nanocubes coexist in the solution. However, the size of bigger nanocubes can not increase infinitely and depends on the reaction condition such as precursor concentration in reaction solution. From line 2 in Fig. 3, we also can see that there is a potential barrier during the smaller nanocubes grow to bigger nanocubes, which can inhibit the growth a part of smaller nanocubes, which is the reason that when the size of bigger nanocubes increases with increasing the concentration of precursor while the size of smaller nanocubes changes little. The crosspoint of line 1 and line 2 is P, the point P have the highest Gibbs free energy and an interesting area S also can be obtained. Area S is a metastable area and the both sides have higher Gibbs free energy, which can inhibit the growth of smaller nanocubes. If we confine d_1 and d_2 in area S, we can obtain monodisperse nanoparticles easily, which offers us a guide theory to synthesize monodisperse nanoparticles. The dimension of S depended on the crosspoint P, P depended on the surface energy of the product while the surface energy depended on product self and the kind and concentration of precursor.

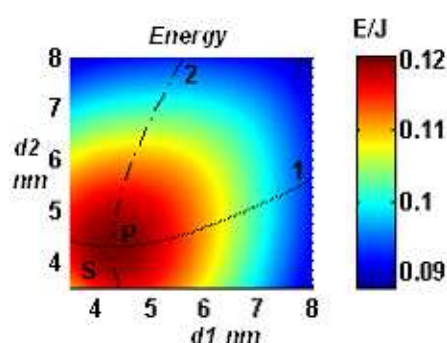


Figure 3 The relationship between Gibbs free energy of the system and particles sizes in system.

Fig.4 shows the effect of surface energy on the crosspoint P. From figure 4a, we can observe that d_1 with a bigger value corresponding to P under lower surface energy, which can obtain a bigger dimension of S, so we can get bigger monodisperse nanoparticles. However, the shift of point P to the right is not unrestraint, when P shifts to the right the energy barrier reduces (as shown in Fig.4b), so more and more smaller nanoparticles can over the barrier to become bigger nanoparticles. The result is only polydispersity particles can be obtained as shown in Fig. 1C. Figure 4c shows the relationship of d_1 and d_2 at point P ($d_1 \approx d_2$).

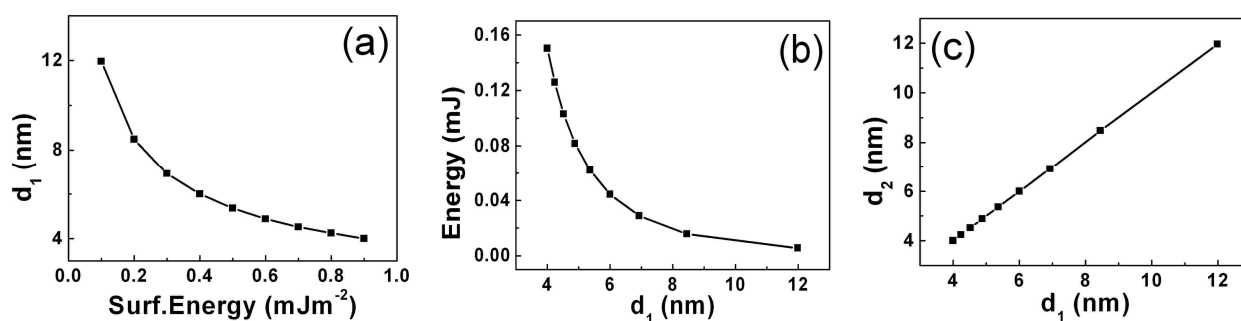


Figure 4 (a) The effect of surface energy on the position of point P; (b) The Gibbs energy at point P changes with d_1 ; (c) The relationship of d_1 and d_2 at point P.

Based on the above discussion we can see that the system appears the most highest Gibbs free energy when $d_1 = d_2$, which can be used to explain it is very difficult to obtain absolutely monodisperse nanoparticles, especially for synthesis bigger monodisperse nanoparticles. In

addition, we studied the effect of temperature and found that temperature also could affect the energy of point P but could not change the position of point P under a specified surface energy. Point P have lower Gibbs free energy under lower temperature, so we can deduce that we can obtain monodisperse nanoparticles more easily in lower temperature.

Conclusions

In summary, GeO₂ nanocubes with different size distribution can be obtained by reverse micelle method. A feasible theory was put forward to explain different sizes nanocubes could coexist in the whole system, which is base on the calculation the free energy of the system. This possible theory can give us helpful guide for synthesis monodisperse nanomaterials in emulsions solution.

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